Year 12 Maths - Triangle geometry

Similar questions - Sine rule, cosine rule and area

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1 In triangle ABC, $AB = 12$ cm, $BC = 9$ cm and angle $BAC = 40°$.

(a) Find the two possible values of angle ACB, to 1 decimal place. **[4]**

(b) Find the length AC in each case, to 2 decimal places. **[4]**

(c) Find the area of the smaller of the two possible triangles. **[2]**

2 In triangle ABC, $AB = 7$ cm, $AC = 10$ cm and angle $BAC = \alpha$. The area of the triangle is 25 cm^2.

(a) Show that $\sin\alpha = \frac{5}{7}$. **[2]**

(b) Find the two possible values of $\cos\alpha$, giving your answers in exact form. **[2]**

(c) Hence find the two possible lengths of BC, to 3 significant figures. **[3]**

Question 1

(a) $\dfrac{\sin C}{12} = \dfrac{\sin 40^\circ}{9}$, so $\sin C = 0.857\ldots$. $C = 59.0^\circ$ or $C = 180^\circ - 59.0^\circ = 121.0^\circ$

(b) $C = 59.0^\circ$: $B = 81.0^\circ$ and $AC = \dfrac{9\sin 81.0^\circ}{\sin 40^\circ} = 13.83$ cm. $C = 121.0^\circ$: $B = 19.0^\circ$ and $AC = \dfrac{9\sin 19.0^\circ}{\sin 40^\circ} = 4.56$ cm

(c) Smaller triangle has $AC = 4.56$: area $= \frac{1}{2} \times 12 \times 4.56 \times \sin 40^\circ = 17.6$ cm^2

Question 2

(a) $\frac{1}{2} \times 7 \times 10 \times \sin\alpha = 25$, so $35\sin\alpha = 25$ and $\sin\alpha = \frac{5}{7}$

(b) $\cos^2\alpha = 1 - \frac{25}{49} = \frac{24}{49}$, so $\cos\alpha = \pm\dfrac{2\sqrt{6}}{7}$

(c) $BC^2 = 49 + 100 - 140\cos\alpha = 149 \mp 40\sqrt{6}$. $BC = \sqrt{149 - 40\sqrt{6}} = 7.14$ cm or $BC = \sqrt{149 + 40\sqrt{6}} = 15.7$ cm